The Islamia University of Bahawalpur
Department of Information Technology

INFT-4041 · BSIT · 5th Semester · Fall 2026

E-Commerce Application Development

Chapter

2

Business Models and Revenue Models

Part I — E-Commerce and Its Models

Taught in	Week 1
Laboratory	Read four real shops; the arithmetic of one order
Assessment	—

This is a class extract. It is *Chapter 2* of the unified handout, printed on its own so that a week's reading is a week's reading. Page numbers restart at 1 and are local to this extract; chapter, figure and table numbers are the ones the full volume gives them, so a reference you follow here lands in the same place there.

Cross-references to other chapters print as their numbers in the full volume. For the whole course in one file — front matter, the HEC tracing table, all twenty-three chapters and all seven appendices — see `ECAD_Fall2026_Handout.pdf`.

Dr. Mustafa Hameed · Department of Information Technology

Generated from `parts/` by `makechapters.py`; do not edit this PDF's source by hand

Chapter 2

Business Models and Revenue Models

Part 1 | Part 2 | Part 3 | Part 4 | Part 5 | Part 6

Part I — E-Commerce and Its Models

Learning Outcomes

By the end of this chapter you will be able to:

- **Distinguish** a business model from a revenue model, and **explain** why a venture can have a sound one and a hopeless other.
- **Classify** an online business by participant — B2C, B2B, C2C, C2B — and say what each classification predicts about the software.
- **Name** the principal revenue models and **identify** which one a given shop is using.
- **Explain** the unit economics of a single order, and **compute** the contribution margin from a set of figures.
- **Argue** which revenue model a given venture should adopt, and **state** the software consequences of that choice.

Before You Start

Chapter 1: the four functions, and the habit of asking which of them a business performs itself.

Key Terms in This Chapter

business model • revenue model • value proposition • B2C • B2B • C2C • C2B • marketplace • commission • subscription • freemium • advertising • affiliate • unit economics • contribution margin • customer acquisition cost • lifetime value • network effect

2.1 Two Questions That Get Confused

Almost every failed online venture answered one of these two questions well and the other not at all.

Business Model and Revenue Model

A **business model** says *what value you create and for whom*: who the customer is, what problem is solved, and why they would choose you rather than the alternative. It is the answer to "why does this deserve to exist?"

A **revenue model** says *how the money reaches you*: who pays, for what, how often, and how much. It is the answer to "who writes the cheque?"

They are independent. The same business model — say, connecting people who want a ride with people who will drive — can be monetised by commission per ride, by a driver subscription, by advertising, or by charging nothing and selling the data. Those four are different companies with different software.

The confusion matters because the two fail differently and the failures look alike from outside. A venture with no business model and a clear revenue model builds something nobody wants and charges correctly for it. A venture with a strong business model and no revenue model builds something people love and cannot pay for it — and it is usually the second that students build.

2.2 Classifying by Who Is at Each End

The oldest classification asks only who is transacting with whom. It is coarse, and it is still the first question worth asking, because it predicts a surprising amount about the software.

Type	Who to whom	What it implies for the software
B2C	A business sells to a consumer	Many small orders, card and wallet payment, a catalogue tuned for browsing, and heavy traffic at unpredictable moments. The default assumption of almost every storefront platform, including the one this course uses
B2B	A business sells to another business	Few large orders, negotiated and customer-specific prices, credit terms rather than immediate payment, purchase orders, approval chains, and quantity breaks. Almost everything in Chapter 10 changes, and most of Chapter 12
C2C	A consumer sells to another consumer	The operator holds no stock and never owns the goods. The hard parts move to trust, dispute resolution and holding money in escrow — Chapter 3 is about this
C2B	A consumer sells to a business	Freelance marketplaces, stock photography, influencer deals. The "product" is a person's time or licence, which breaks the stock model entirely

Why the B2B/B2C split is the one that bites

A platform built for B2C assumes one price per product. B2B does not have one price per product — it has a price per product *per customer*, often negotiated, often with quantity breaks, sometimes expiring.

This is the most common way a student project, or a real small business, picks the wrong tool. Everything looks fine until the first wholesale customer asks for their agreed rate, and the answer turns out to require a different data model rather than a setting. Chapter 4 makes you ask this question before you choose, which is the only time asking it is cheap.

2.3 The Revenue Models

There are not very many, and most businesses combine two or three.



Figure 2.1: The five revenue models a shop can use, and the single fact about each that most shapes the software. Most real businesses run two of them at once.

2.3.1 Sales margin

You buy at one price and sell at another. The oldest model, and the one the storefront software in this course is built around by default.

Its distinguishing feature, from a software point of view, is that *you hold stock*. Stock means a quantity that can be wrong, can be reserved, can go negative under concurrency, and must be reconciled against physical reality. Chapter 6 and Chapter 18 spend real effort on this.

2.3.2 Commission

Somebody else sells; you take a percentage for having made it possible. This is the marketplace model, and it is attractive because you hold no stock and your costs barely rise with volume.

What you take on instead is everything in Chapter 3: two parties who do not trust each other, money that must be held until both are satisfied, and disputes that someone has to adjudicate. You have also acquired a problem the sales model does not have — you must attract *both* sides, and neither arrives before the other.

2.3.3 Subscription

The customer pays regularly for continued access or delivery. Revenue becomes predictable, which is why investors like it and why so much of the web has moved to it.

The software consequence is recurring billing, which is substantially harder than one-off payment: cards expire, payments fail silently, customers pause and resume, and the price may change mid-contract. Chapter 14 explains why a stored card is a different kind of integration from a single charge.

2.3.4 Advertising and affiliate

Advertising sells the attention of your visitors to somebody else. It needs large traffic before it pays anything at all, which makes it a poor first revenue model and a common fantasy.

Affiliate revenue is the mirror image: you send a visitor to somebody else's shop and are paid if they buy. Cheap to start, and it leaves you without a customer — you never learn who bought, cannot serve them again, and the relationship belongs to the merchant. Chapter 21 is where this becomes a strategy rather than an accident.

2.4 The Arithmetic of One Order

A revenue model is a claim that can be checked, and the check is simpler than students expect. Take one order and follow the money out of it.

Worked Example: Does this order make money?

A shop sells a pair of shoes at **Rs. 4,500**, delivered anywhere in Pakistan, with free returns. Work out what is left.

Step 1 — what came in.
Revenue per order: **Rs. 4,500**.

Step 2 — the cost of the thing itself.
The shoes cost the shop Rs. 2,800 to buy.
Remaining: $4{,}500 - 2{,}800 =$ **Rs. 1,700** (a gross margin of 38%).

Step 3 — the cost of taking the money.
The gateway charges 2.5% plus Rs. 15 per transaction: $0.025 \times 4{,}500 + 15 = 112.5 + 15 =$ Rs. 127.50.
Remaining: $1{,}700 - 127.50 =$ **Rs. 1,572.50**.

Step 4 — the cost of delivery.
Courier, Rs. 250, absorbed by the shop because delivery was advertised free.
Remaining: $1{,}572.50 - 250 =$ **Rs. 1,322.50**.

Step 5 — returns, which are not optional.
20% of shoe orders come back. A return costs the courier fee again (Rs. 250) and the item cannot always be resold; assume a further Rs. 200 of loss on average. Expected cost per *order placed*: $0.20 \times (250 + 200) =$ Rs. 90.
Remaining: $1{,}322.50 - 90 =$ **Rs. 1,232.50**.

Step 6 — the cost of finding the customer.
The shop spends Rs. 60,000 a month on advertising and gets 50 orders from it: a **customer acquisition cost** of $60{,}000/50 =$ Rs. 1,200.
Remaining: $1{,}232.50 - 1{,}200 =$ **Rs. 32.50**.

The answer. The order that looked like a 38% margin earns **Rs. 32.50**, which is **0.7%** of the sale price — and that is before rent, salaries, hosting or tax. One extra return, or a 3% gateway instead of 2.5%, and it is a loss.

What this is really telling you. The two numbers that decided it were *returns* and *acquisition cost*, and neither appears anywhere in the product's price. A shop that measures only gross margin believes it is making 38%. This is why Chapter 17 is careful about which numbers on a dashboard are worth acting on.

Contribution Margin, CAC and LTV

Contribution margin is what one order contributes after all the costs that vary with that order — goods, payment fees, delivery, expected returns. It is what is left to pay the costs that do not vary.

Customer acquisition cost (CAC) is the marketing spend divided by the customers it produced.

Lifetime value (LTV) is the total contribution margin a customer produces before they stop buying.

The test a business must pass is LTV > CAC, by a comfortable multiple. The worked example above passes only if the customer orders again — its CAC is nearly its whole first-order margin, which is survivable for a business with repeat customers and fatal for one without.

Where the network effect comes from, and where it does not

A **network effect** exists when the service gets better for each user as more users join. A marketplace has one: more sellers make it more useful to buyers, which attracts more sellers.

A shop selling its own goods has *no* network effect. Your thousandth customer makes the shop no better for your first. Volume may buy you better purchase prices, which is a real advantage, but it is not a network effect and it does not protect you the same way.

This matters because the two justify completely different strategies. Running at a loss to grow is rational when a network effect will eventually make you defensible, and is just losing money when it will not. Plenty of ventures have adopted the first strategy while having the second business.

Common Pitfalls

- **Naming a business model and thinking you have named a revenue model.** "We are a marketplace for tutors" says nothing about who pays.

- **Quoting gross margin as if it were profit.** The worked example went from 38% to 0.7% without anything unusual happening.
- **Forgetting returns.** In clothing and footwear they run at 20–40% and they are the difference between a viable shop and a dead one.
- **Advertising as a first revenue model.** It pays at scale and nothing before it, so a new venture choosing it has chosen to earn nothing for a long time.
- **Assuming the platform is model-agnostic.** It is not. A B2C storefront has one price per product, and discovering that during implementation is expensive — see Chapter 4.
- **Treating "free delivery" as marketing.** It is a cost of Rs. 250 per order that appears in no price list and is paid out of the margin.

Try It Yourself: The model behind the shop you will build

Still no software — Chapter 5 starts that. This laboratory produces the one-page brief that Chapter 4 turns into a decision and Chapter 23 is eventually marked against, so it is worth real effort.

Choose a shop to build. It must be plausible, small, and local enough that you can check your assumptions: a bookshop, a bakery, a tailor, a spare-parts dealer, a plant nursery. Do not choose "an Amazon competitor".

Then produce, on one page:

1. **The business model in two sentences.** Who is the customer, what problem is solved, and why you rather than the shop down the road or the marketplace they already use.
2. **The classification**: B2C, B2B, C2C or C2B — and if you are tempted to say "both", say which one the *software* must assume, because it can only assume one.
3. **The revenue model**, from Figure 2.1, with the reason for choosing it over the next most plausible one.
4. **The arithmetic of one order**, following the six steps of the worked example with your own figures. Find real numbers where you can: gateway fees and courier rates are published, and a shopkeeper will usually tell you their return rate if you ask politely. Mark every number you guessed.
5. **The two numbers that decide it.** From your own arithmetic, name the two inputs the result is most sensitive to, and say what would have to be true for the shop to lose money.

What to notice. Almost everyone finds that the contribution margin is far thinner than they expected, and that the deciding numbers are not the ones they thought they were building a business around. That discovery is the point of the exercise, and it is much cheaper to make here than in Chapter 23.

Checkpoint — Test Yourself

1. A food delivery app charges restaurants 20% commission, charges customers a delivery fee, and sells promoted placement in search results. Name the business model and all the revenue models, and say which single one you would expect to be most profitable per unit.
2. A shop has a contribution margin of Rs. 400 per order and a customer acquisition cost of Rs. 1,500. Under what condition is this business viable? State it as an inequality and say what would need measuring.
3. Explain why a B2B shop cannot usually be built by configuring a B2C platform, naming the specific assumption that breaks.

Chapter Summary

- A business model says what value is created and for whom; a revenue model says who pays. They are independent, and a venture can have a good one and a fatal other.
- B2C, B2B, C2C and C2B classify by who transacts. The split that most affects the software is B2B against B2C: B2B has a price per customer, not a price per product.
- The five revenue models are sales margin, commission, subscription, advertising and affiliate. Holding stock or not is the fact about each that most shapes the system you have to build.
- Follow one order through six deductions — goods, payment fee, delivery, returns, acquisition — and the margin usually collapses. In the worked example, 38% became 0.7%.
- The viability test is LTV > CAC by a comfortable multiple, which for a thin first-order margin means repeat custom is not optional.
- Marketplaces have network effects; shops selling their own goods do not, and strategies that assume one are expensive when it is absent.

Review Questions

1. (MCQ) "We connect home cooks with office workers who want lunch." This is: (a) a revenue model; (b) a business model; (c) both; (d) neither.

2. (MCQ) Which revenue model most changes the *payment* integration a shop needs? (a) Sales margin; (b) affiliate; (c) subscription; (d) advertising.

3. (Short) Distinguish gross margin from contribution margin, and give two costs that appear in one and not the other.

4. (Short) Explain why a marketplace has a harder launch than a shop, despite holding no stock.

5. (Short) A clothing shop reports 45% gross margin and is losing money. Give the two most likely explanations, and say what you would measure first.

6. (Applied) A shop sells a product at Rs. 2,000 that costs it Rs. 1,100. Gateway: 2.9% + Rs. 20. Delivery: Rs. 200, charged to the customer at Rs. 150. Returns: 12%, each costing Rs. 200 in courier and Rs. 150 in unsellable stock. Compute the contribution margin per order, showing each step. Then find the return rate at which the shop breaks even.

7. (Applied) Take the venture from this chapter's laboratory and argue for a *different* revenue model than the one you chose. Give the strongest case you can, then say which software decisions in Parts III and IV would have to change if you adopted it.